Assessing the Climatological Significance of Cloud Seeding in the Western United States

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Abstract

Cloud seeding has been regularly practiced in the United States since the 1940's. While targeted research field campaigns indicate that cloud seeding can produce local increases in rainfall, there has been no previous large-scale assessment of the effectiveness of operational cloud seeding as it has been practiced in the United States. To address this gap, we leverage the first large-scale, publicly available data set of cloud seeding activities in the Western United States (Donohue and Lamb, 2025), compiled from reports filed under the US Weather Modification Reporting Act of 1972. Leveraging these historical cloud seeding records, we analyzed 823 distinct operational periods from 2000 – 2025, corresponding to seeding at 139 distinct seeding sites. Using the reported locations and times of historical cloud seeding operations, along with historical datasets of precipitation, snowpack, and streamflow, we analyze whether cloud seeding has had a climatologically significant impact on regional precipitation or snowpack. By comparing seeded sites to nearby, and climatologically similar, unseeded sites, we find little evidence that operational cloud seeding in the United States has left a climatologically significant signature. However, we identify opportunities to improve weather modification and geoengineering reporting going forward to enhance the measurement of cloud seeding efficacy, despite limited federal reporting requirements existing today.

Keywords: Weather Modification, Causal Inference, Climatological Impacts, Geoengineering, Environmental Policy

1 Introduction

Weather modification refers to methods used to augment rainfall on short time scales in order to reduce the impacts of drought or to increase precipitation for agriculture.

While past research field campaigns such as the SNOWIE project quantified the impact of cloud seeding locally and indicated that snowfall can be significantly increased by cloud seeding[1], there has been little attempt to provide a broad-scale climatological analysis of cloud seeding as it has been used operationally in the United States. Past efforts to study the climatological effectiveness of cloud seeding have typically focused on single sites or watersheds.

For example, the Wyoming Weather Modification Pilot Program (WWMPP) program used a randomized experimental approach to study whether increased snowfall from orographic cloud seeding could be detected, by randomly seeding either the Medicine Bow or Sierra Madre Mountain ranges over 6 winter seasons (2008-2014) [2]. One mountain range served as the control (unseeded case), while the other range was seeded. The statistical analysis of 154 randomized seeding experiments was inconclusive however, likely due to potential unintended downwind impacts of seeding on the Medicine Bow range by seeding over the Sierra Madre, and insufficient amounts of silver iodide used during seeding.

To provide a broad-scale analysis of the climatological impacts of cloud seeding in the US, we instead focus on large-scale analysis of historical federal cloud seeding records reported under the 1972 Weather Modification Reporting Act. We leverage a recently developed historical data set of cloud seeding activities in the United States[3]. This data set was developed by applying optical character recognition and OpenAI's o3 large language model to extract information about the project name, year, season, state, operator, seeding agent, apparatus used for deployment, stated purpose, target area, control area, start date and end data of all publicly reported cloud seeding activities. This method was validated through the performance of the data extraction pipeline on a manually labeled subset of 200 records, achieving an average accuracy of 98.38% across all fields. This structured data set, encompassing 823 distinct operational periods from 2000 – 2025, represents the first large-scale, publicly available data set of cloud seeding activities for the US that can facilitate large-scale historical analysis.

Using this data set, we use causal inference, a statistical method to estimate the effect of an intervention, to understand whether cloud seeding is an effective strategy for augmenting snowpack and precipitation from a climatological perspective. Using the reported locations and times of historical cloud seeding operations, along with historical datasets of precipitation and snowpack, we analyze whether cloud seeding as it has been used operationally in the United States has had any climatologically significant impact in augmenting precipitation and snowpack regionally. In addition, we examine streamflow gauge measurements to determine whether watersheds associated with runoff from regions with cloud seeding have shown measurable increases in waterflow. Our approach makes it possible to rigorously evaluate cloud seeding effectiveness on a climatological scale across the US for the first time.

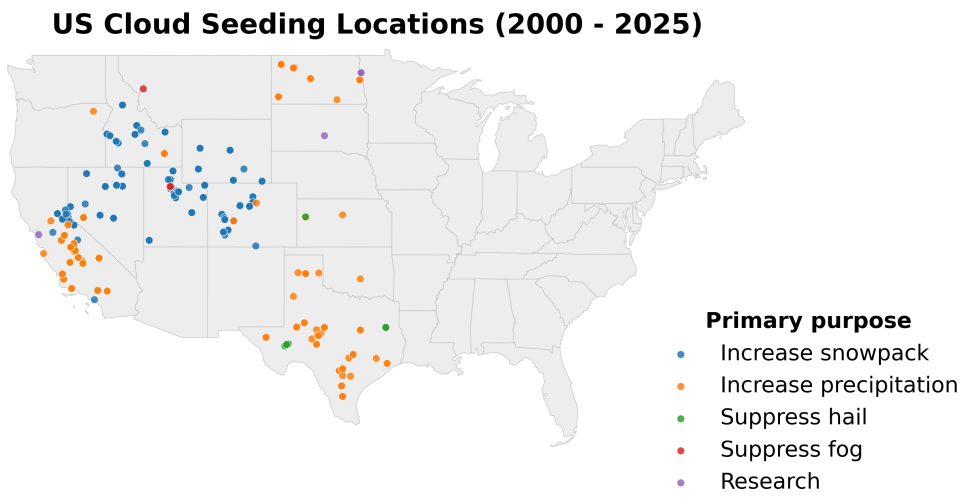

Fig. 1 Locations of cloud seeding in the Western U.S., colored by the primary stated purpose.

2 Results

Through analysis of the historical cloud seeding data set, we identified 139 unique cloud seeding locations over the 25 year record (Figure 1). The Supplemental Information provides detailed tables for each state, giving the approximate latitude and longitude location, site ID number, years of operation, seasons, project names, and operators. Texas had the most unique sites (33), while California had 29 unique sites, and Colorado and Utah both had 15 identified unique sites. Figure 2 shows the identified cloud seeding sites from the historical data set, and the periods during which they were seeded each year from 2000 to 2025. The majority of cloud seeding sites were regularly seeded during several months of each year, with 91 out of the 139 sites active more than a single year, and a mean of 5.6 years of seeding per active site. Table 1 provides the 10 most active sites, with the number of active years. For example, the Eastern San Juan Mountains Program in Colorado reported seeding in 24 out of the 25 years in our record. This demonstrates that many regions of the Mountain West regularly rely on cloud seeding to increase water reservoirs and mountain snowpack. Despite long-term cloud seeding programs at many locations, there are still significant disagreement about whether cloud seeding is an effective strategy for augmenting precipitation.

To understand whether cloud seeding has had any measurable impact on precipitation patterns, we estimate the causal effect of cloud seeding on precipitation using a Difference-in-Difference (DiD) framework.

Because seeding sites are not climatologically similar to one another, comparing precipitation differences between periods with seeding or no seeding cannot be used

Table 1 Ten most active weather modification sites (2000-2025), ranked by number of active years.

Site ID	Years	Project	Lat	Lon	State
33	24	Eastern San Juan Mountains Cloud Seeding Program	38.07	-107.45	Colorado
40	22	Central Colorado Program	39.60	-106.52	Colorado
18	21	Upper Tuolumne River Weather Modification Project	37.74	-120.28	California
21	21	Mokelumne, Lake Almanor	38.37	-119.53	California
115	21	PGCD Weather Modification Program	35.35	-101.38	Texas
125	20	Alta/Snowbird Ski Resort Cloud Seeding Program	40.58	-111.66	Utah
6	18	Kern River Cloud Seeding Program	35.98	-118.84	California
12	18	Kings River Cloud Seeding Program	36.73	-119.50	California
14	18	Mokelumne Cloud Seeding Program	36.99	-119.79	California
32	18	Telluride Ski Area Upper San Miguel/San Juan Basin Cloud Seeding Program	37.94	-107.81	Colorado

as a way to assess climatological effectiveness, because sites with more wintertime precipitation are more likely to be seeded in the winter. In addition, looking at relative mean differences between years at the same site that are seeded or not seeded is also potentially problematic, because the decision to seed in any given year is not independent from the observed precipitation. In addition, because many of the sites are seeded during the same months every year, there are often very few control periods for specific months.

A better approach to understand the climatological impact of cloud seeding is to instead compare precipitation in the same periods at climatologically similar sites that were not seeded. NOAA's 17-4 form asks for operators to identify both target and control sites; the control site is a site that is not targeted for weather modification activities but can be used as a climatological control. However, in practice, less than half of operators actually provided control sites. In the records, we found that only 377 out of the 832 reports (45%) identified control sites.

Recent observations from the Surface Atmosphere Integrated Field Laboratory (SAIL) campaign in the East River Watershed (part of the Upper Colorado River Basin) indicated that cloud seeding can contribute to measurable increases in ice nucleating particle (INP) concentrations in regions where seeding regularly occurs [4]. This suggests a mechanism by which cloud seeding can have an impact on regional precipitation and snowfall, even several weeks after operational cloud seeding. Wind-blown silver iodide particles could also act as INP in the larger region (beyond the specific location of the cloud seeding site), impacting snowpack downwind from identified seeding sites.

- Separate analysis by "snowpack" vs. "rainfall" augmentation and assess each separately. Remove the other weather modification activities (e.g. fog and hail suppression, geo-engineering studies)

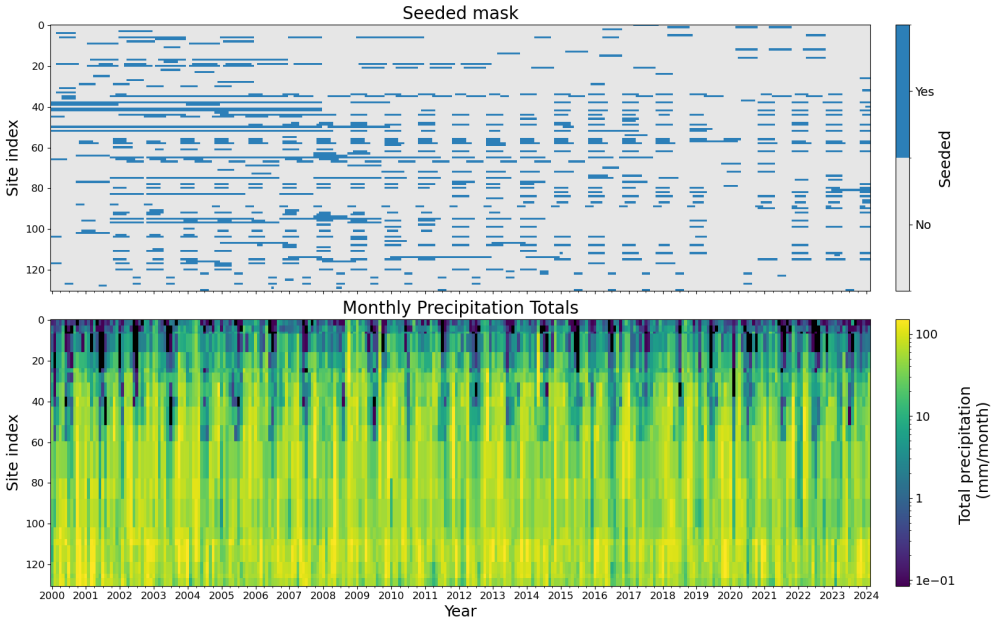

Fig. 2 Overview of cloud seeding on a per site level (top) and the corresponding monthly precipitation totals (bottom) at the latitude and longitude location of each site.

3 Methods

Identification of Seeded Sites

We use the `Target` and `State` fields in the cloud seeding data set, geo-coded with the Google Maps API[5] to get latitude and longitude locations of seeding sites. Based on these latitude and longitude locations (truncated to 2 decimal points), we determined that there were 139 unique seeding sites over the 25 year record.

Identification of Control Sites

When provided by the cloud seeding operator, we use the stated `Control Area` as the comparison site. We use the `Control` and `State` fields in the cloud seeding data set, geo-coded with the Google Maps API to get latitude and longitude locations for the control sites.

Alternatively, we apply the K-Nearest Neighbors algorithm to learned embeddings of seasonally resolved climatological factors and atmospheric composition to identify sites that are climatologically similar to the target seeding sites [**?**].

- Use a transformer architecture and learn embeddings that can predict precipitation; the attention mechanism can determine how we should weight the various climatological and atmospheric composition factors at each latitude and longitude point.
- Climatological factors: elevation, terrain ruggedness

- Vegetation, soil moisture (from ERA5-Land surface variables)
- GridMET (1979 - present, 4 km resolution, hydrological modeling in the Western US). https://www.climatologylab.org/gridmet.html. Climate variables: Maximum temperature, minimum temperature, precipitation accumulation, downward surface shortwave radiation, wind-velocity, humidity (maximum and minimum relative humidity and specific humidity; Derived variables: Reference evapotranspiration (ASCE Penman-Montieth), Energy Release Component*, Burning Index*, 100-hour and 1000-hour dead fuel moisture, mean vapor pressure deficit, 10-day Palmer Drought Severity Index *fuel model G (conifer forest)
- Mean temperature, mean RH, lower tropospheric stability, wind speed and direction, integrated vapor transport (IVT)
- Background aerosols: AERONET (AOD, dust, smoke); we use the AEROMAP data sets [6] for compilations of PM2.5 and PM10 aerosol concentrations at the surface. AEROMAP data sets provide annual and monthly climatological information for coarse and fine model aerosol species at 1 degree by 1 degree (Al, BC, NH4, NO3, Na, OM, PM, SO4) from Community Atmosphere Model version 6 (CAM6) simulations. Some of the aerosol species are from CAM-CHEM simulations. Double-check the details in the paper (https://zenodo.org/records/15391280).
- Latitude and longitude location. Distance from seeding sites, but limit how close to actual seeding sites these points can be (to e.g. remove any possibility of wind-blown impacts of seeding aerosols on neighboring sites). First identify 2000 nearest neighbors, and then remove points that are too close or too far away based on latitude and longitude location.
- Find 100 nearest neighbors for each site and then use boot-strapping over these 100 to estimate uncertainty.

Estimating the Causal Effect of Historical Cloud Seeding on Precipitation

We estimate the causal effect of cloud seeding on precipitation using a difference-in-differences (DiD) design for seeding programs that initiate during the 2000–2025 observation window. Restricting attention to these programs provides an untreated pre-intervention period within the available data, enabling credible causal estimation using DiD.

For each site meeting this criterion, we define the treatment start as the first year in which seeding activity is reported and compare changes in precipitation before and after treatment between the seeded target area and a nearby unseeded control area. This approach measures relative changes over time, differencing out time-invariant spatial heterogeneity under the assumption of parallel trends. To assess the plausibility of the parallel trends assumption, we examine differential precipitation trends in the years immediately preceding seeding to ensure that treated and control areas exhibit similar precipitation dynamics prior to treatment.

For sites that are seeded throughout the 2000–2025 record and therefore lack a clearly defined untreated pre-period, DiD estimation is not feasible. For these locations, we report precipitation differences between the target and control areas as descriptive comparisons of climatological signatures rather than causal effect estimates. These

descriptive contrasts should be interpreted more cautiously, as treatment is endogenous and the counterfactual untreated outcomes are not observed.

All causal estimates are therefore identified from sites with observed seeding initiation within the study window, while long-running programs contribute to the analysis through descriptive comparison.

Code to reproduce this analysis is available at https://github.com/jdonohue44/HistoricalCloudSeedingDiffDiff.

Precipitation data sets

Here we use NOAA Climate Data Record of GPCP Satellite-Gauge Combined daily gridded product selected at the latitude and longitude location of the 139 unique seeding sites.

- NOAA Global Precipitation Climatology Product (GPCP) one-degree daily precipitation product provides daily, global 1x1 degree gridded fields of precipitation totals from Oct. 1996 to present []. Version 1.3. (https://www.ncei.noaa.gov/products/global-precipitation-climatology-project) Can potentially underestimate orographic enhancements of precipitation; alternatively, can overestimate precipitation in cases of orographic enhancement of ice formation at the top of storms that lead to inefficient rainfall. PMW schemes may also struggle over snowy/frozen surfaces, so estimates of cold-season precipitation over land can be problematic.
- NASA's Integrated Multi-satellitE Retrievals for GPM (IMERG) data product. This fuses data from the TRMM satellites (1998-2015) and the more recent GPM missions (2014-present) to provide a continous global precipitation data set over two decades. 0.1 degree (10 km resolution, 30 min. resolution)
- PRISM Climate Group data set (Daily, 1981 - present, 800 m - 4 km, monthly and daily records available). Good in the Western US. Cloud Optimized GeoTIFF datasets.
- Stage IV Precipitation data set (Gauge-corrected radar, 4 km resolution, hourly since 2002).
- For snow-water equivalent: SNODAS (1 km resolution, daily). SNOTEL (specific stations?)
- Physics-informed ML data product from Pierre's group [7]

Stream-flow measurements

We use basin level information from the CAMELS data set to determine whether streamflow gauge changes where climatologically significant.

- Identify the stream basin associated with sites
- Determine independent stream basins to compare with (for climatologically similar sites?)
- Use one of the ML models for CAMELS

Long-term monitoring of streamflow impacts from cloud seeding has been identified as a key indicator of effective increases in water resources [8], although there have been few past studies that have addressed this issue.

Establishing the Limitations of Current Federal Reporting

To estimate whether current federal reporting requirements would allow for climatological impacts of cloud seeding to be assessed, we ...

- Use OCR to get number of seeded days (or just focus on one representative site)
- Increase precipitation by 15-30% across seeded sites on X randomly selected days of seeding in a month (does this lead to statistically significant changes in precipitation when compared with the control sites and including measurement uncertainty?)
- What level of detail would be needed? E.g. if we knew which specific days, would this be enough? Or do we need even more stringent reporting to provide this level of detail?

4 Discussion

This analysis points to limitations in the current federally mandated requirements for cloud seeding. There is currently a lack of detailed information about the specific location and timing of cloud seeding activities. Current reports provide information about the total estimated number of days per month that weather modification activities are performed, without specific details about the dates or hours. In addition, locations provided are generally very vague (given place name or county), rather than providing a specific latitude and longitude location or flightpath (in the case of airborne cloud seeding). It is not always clear which planned activities actually took place, as often operational decisions are made in response to meteorological conditions, and final reporting does not require an assessment of the specific times in which seeding actually did occur. Finally, the amount of materials deployed is often stated without uncertainty or indication of how these materials were measured. There is also inherently some limitations in terms of using self-reported cloud seeding activities which are not independently verified in other ways, which could lead to inaccurate or missing records.

As geoengineering approaches becomes more broadly considered as a policy option, it is also important to consider reporting requirements needed to assess the effectiveness of various intervention strategies. Measurement methodologies for monitoring intervention strategies also need to be assessed scientifically to determine whether they can provide sufficient spatial and temporal resolution to infer the effectiveness of various strategies, given current limitations in both models and satellite observations (cite e.g. the recent paper on marine cloud brightening, recent SAI efforts, etc.) For example, high resolution radar coupled with precipitation gauge information can provide detailed information about regional responses to cloud seeding[1]; such measurements might be more broadly integrated into federal reporting requirements. In addition, while we have focused here on traditional weather modification approaches, federal reporting of climate geoengineering currently is mandated under the same 1972 law, and the unique physical and chemical processes and larger climate response at play for these approaches need to be considered when developing federal reporting requirements for these methods, unique from cloud seeding.

5 Conclusion

In this paper, we have examined whether operational cloud seeding as it has been used in the United States over the past 25 years has had any significant impacts on precipitation or snowfall regionally.

Our research also points to limitations in current federal reporting requirements.

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Declarations

Some journals require declarations to be submitted in a standardised format. Please check the Instructions for Authors of the journal to which you are submitting to see if you need to complete this section. If yes, your manuscript must contain the following sections under the heading 'Declarations':

- Funding: Zegar Family Foundation, Columbia Data Science Institute
- Conflict of interest/Competing interests (check journal-specific guidelines for which heading to use)
- Ethics approval and consent to participate
- Consent for publication
- Data availability:
- Materials availability
- Code availability: Python code for the climatological analysis is available at our public GitHub repository [**?**]
- Author contribution

If any of the sections are not relevant to your manuscript, please include the heading and write 'Not applicable' for that section.

References

[1] Friedrich, K., Ikeda, K., Tessendorf, S.A., French, J.R., Rauber, R.M., Geerts, B., Xue, L., Rasmussen, R.M., Blestrud, D.R., Kunkel, M.L., Dawson, N., Parkinson, S.: Quantifying snowfall from orographic cloud seeding. Proceedings of the National Academy of Sciences **117**(10), 5190–5195 (2020) https://doi.org/10.1073/pnas.1917204117 . Accessed 2026-02-09

[2] The Wyoming Weather Modification Pilot Program, Level II Study, Executive Summary. https://wwdc.state.wy.us/weathermod/WYWeatherModPilotProgramExecSummary.html Accessed 2026-02-18

[3] Donohue, J.J., Lamb, K.D.: Structured dataset of reported cloud seeding activities in the United States (2000–2025) using an LLM. Scientific Data **12**(1), 1996 (2025) https://doi.org/10.1038/s41597-025-06273-1 . Accessed 2026-02-09

[4] Zhou, R., Perkins, R., Juergensen, D., Barry, K., Ayars, K., Dutton, O., DeMott, P., Kreidenweis, S.: Seasonal variability, sources, and parameterization of ice-nucleating particles in the Rocky Mountain region: importance of soil dust and biological contributions. Atmospheric Chemistry and Physics **26**(2), 1515–1535 (2026) https://doi.org/10.5194/acp-26-1515-2026 . Accessed 2026-02-11

[5] Google Maps Platform Documentation | Geocoding API. https://developers.google.com/maps/documentation/geocoding Accessed 2026-02-12

[6] Mahowald, N.M., Li, L., Vira, J., Prank, M., Hamilton, D.S., Matsui, H., Miller, R.L., Lu, P.L., Akyuz, E., Meidan, D., Hess, P., Lihavainen, H., Wiedinmyer, C., Hand, J., Alaimo, M.G., Alves, C., Alastuey, A., Artaxo, P., Barreto, A., Barraza, F., Becagli, S., Calzolai, G., Chellam, S., Chen, Y., Chuang, P., Cohen, D.D., Colombi, C., Diapouli, E., Dongarra, G., Eleftheriadis, K., Engelbrecht, J., Galy-Lacaux, C., Gaston, C., Gomez, D., González Ramos, Y., Harrison, R.M., Heyes, C., Herut, B., Hopke, P., Hüglin, C., Kanakidou, M., Kertesz, Z., Klimont, Z., Kyllönen, K., Lambert, F., Liu, X., Losno, R., Lucarelli, F., Maenhaut, W., Marticorena, B., Martin, R.V., Mihalopoulos, N., Morera-Gómez, Y., Paytan, A., Prospero, J., Rodríguez, S., Smichowski, P., Varrica, D., Walsh, B., Weagle, C.L., Zhao, X.: AERO-MAP: a data compilation and modeling approach to understand spatial variability in fine- and coarse-mode aerosol composition. Atmospheric Chemistry and Physics **25**(9), 4665–4702 (2025) https://doi.org/10.5194/acp-25-4665-2025 . Accessed 2026-02-12

[7] Zhao, W., Fang, J., Yang, T., Lian, X., Winkler, A.J., Sun, F., Gentine, P.: Observation-constrained physical snow water equivalent simulations using a physics-guided machine learning approach. Accessed 2026-02-12

[8] Tessendorf, S.A., Boe, B., Geerts, B., Manton, M.J., Parkinson, S., Rasmussen, R.: The Future of Winter Orographic Cloud Seeding: A View from Scientists and Stakeholders. Bulletin of the American Meteorological Society **96**(12), 2195–2198 (2015) https://doi.org/10.1175/BAMS-D-15-00146.1 . Chap. Bulletin of the American Meteorological Society. Accessed 2026-02-18